The Insolation

*The only source of energy for the Earth's atmosphere and its surface is the Sun. The outer layer of the Sun i.e. the photosphere releases energy at a constant rate. As a result, each per unit area of the Earth also receives constant solar energy. This phenomenon is known as the **solar constant**.

* Distribution of Solar Energy – Normally the solar energy received **decreases from** the **equator towards poles.** However, there are temporal changes in the energy received at every latitude. Since the Sun's position in comparison to that of the Earth shifts northwards and southwards during Earth's revolution due to Earth's tilta the maximum solar energy received also shifts north and south of the equator in the same respect. 20° North Latitude receives the maximum solar energy not the equator.

*Regions where the Sun's rays fall in a slanted manner receive less solar energy compared to regions where the Sun's rays are falling vertically.

***Albedo** is the measure of how much light that hits a surface is reflected without being absorbed. Earth has an average albedo of **30%** whereas the maximum albedo effect is produced by surfaces such as **glaciers and snow (75-95%)**. **Deserts** have an albedo of **20-30%**, **paddy fields 3-15%** and **prairies 10-20%**.

*Water vapors are capable of absorbing the infrared radiation waves and the outgoing long waves of the dispersed solar waves however, water vapors are transparent to the ultraviolet rays of the sun, hence are incapable of absorbing them. * On clouded nights the amount of water vapours in the atmosphere is greater compared to a clear sky night. The outgoing long waves are absorbed by the water vapours present in the atmosphere. Hence, cloudy nights are warmer than clear sky nights.

*Heat Budget-Energy Balance – the rays of the Sun enter the Earth's atmosphere as short wave solar radiation and Escape the Earth as long-wave terrestrial radiation. The balance between incoming solar radiation and outgoing terrestrial radiation is known as Heat Budget – Energy Balance.

*Long-wave terrestrial radiation plays a major role in keeping the atmosphere warm. The atmosphere is warmed primarily by the difference between incoming solar radiation and outgoing terrestrial radiation. This warmth is captured by gases like carbon dioxide leading to the green house effect and global warming.

1. Assertion (A) : The atmosphere receives most of the heat only indirectly from the sun and directly from the earth's surface.

Reason (R) : The conversion from shortwave solar to long wave terrestrial energy takes place at the earth's surface.

Code:

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P.P.C.S. (Pre) 1997

Ans. (a)

The principal source of atmospheric and Earth's surface energy is the Sun. Insolation is the incoming solar energy intercepted by the Earth. The energy radiated from the Sun is known as short-wave solar radiation whereas the energy radiated from Earth is known as long-wave radiation.
The long-wave radiation is absorbed by the atmospheric gases particularly by carbon dioxide and the other greenhouse gases. Thus, the atmosphere is indirectly heated by the Earth's radiation. Increasing concentrations of greenhouse gases such as carbon dioxide and methane increase the temperature of the lower atmosphere by restricting the outward passage of emitted radiation, resulting in global warming.

2. Normally, the temperature decreases with the increase in height from the Earth's surface, because

1. The atmosphere can be heated upwards only from the Earth's surface.
2. There is more moisture in the upper atmosphere.
3. The air is less dense in the upper atmosphere.

Select the correct answer using the codes given below :

(a) Only 1 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

I.A.S. (Pre) 2012

Ans. (c)

Globally, insolation is the main factor behind the distribution of temperature. By absorbing Sun rays, the surface of Earth gets warmer. This is the reason why temperature decreases as height increases. At high altitudes air density is less with little humidity. Hence, statement (1) and (3) are correct while (2) is wrong. Therefore, option (c) is the correct answer.

3. **Which one of the following reflects more sunlight as compared to the three?**
   (a) Sand desert
   (b) Paddy cropland
   (c) Land covered with fresh snow
   (d) Prairie land

I.A.S. (Pre) 2010

Ans. (c)

Albedo is the fraction of solar energy (shortwave radiation) reflected from the Earth back into space. It is a measure of the reflectivity of the Earth's surface. Land covered with fresh snow has a high albedo (75-95%), most sunlight hitting the surface bounces back towards space. The average albedo of Earth is the 30%. The sand desert has an albedo of 20-30%. Paddy cropland has an albedo 3-15% and prairie land has an albedo 10-20%.

4. **Read the following statements and choose the correct option :**
   1. **Moving away from the sea-force of tropical cyclones decreases.**
   2. **Tropical cyclone gets its energy from release of latent heat on account of condensation of moisture that wind gathers after moving over the ocean which decreases as it moves away from sea.**

   (a) 1 and 2 both are true, and 2 is correct reasoning of 1.
   (b) 1 and 2 both are true, but 2 is not correct reasoning of 1.
   (c) 1 is true, but 2 is false.
   (d) 1 and 2 both are false.

Chhattisgarh P.C.S. (Pre), 2022

Ans. (a)

Moving away from the sea, force of tropical cyclones decreases because tropical cyclones get their energy from release of latent heat on account of condensation of moisture that wind gathers after moving over the ocean which decreases as it moves away from sea.

5. **Which one of the following is associated with 'Albedo'?**
   (a) Transmitting power (b) Absorbing power
   (c) Emissive power (d) Reflecting power

U.P. P.C.S. (Pre) 2019

Ans. (d)

Albedo is the fraction of light that a surface refllects. Significantly frozen ice will have a high albedo, which means it reflects solar radiation into space, while green areas such as forests and plains will have a low albedo.

6. **Consider the following statements:**
   1. **The annual range of temperature is greater in the Pacific Ocean than that in the Atlantic Ocean.**
   2. **The annual range of temperature is greater in the Northern Hemisphere than that in the Southern Hemisphere.**

   Which of the statements given above is/are correct?
   (a) 1 only (b) 2 only
   (c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2007

Ans. (b)

The larger size of the ocean, less the variation in the annual temperature. Therefore, more temperature variation is found in the Atlantic Ocean as compared to the Pacific Ocean.

7. **Which of the following is/are correct inference/ inferences from isothermal maps in the month of January?**
   1. **The isotherms deviate to the north over the ocean and to the south over the continent.**
   2. **The presence of cold ocean currents, Gulf Stream and North Atlantic Drift make the North Atlantic Ocean colder and the isotherms bend towards the north.**

   Select the answer using the code given below :
   (a) 1 only (b) 2 only
   (c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2024

Ans. (a)

The temperature distribution is generally shown on the map with the help of isotherms. The isotherms are lines on the map joining places having equal temperature. In general the effect of latitude on temperature is well pronounced on the map, as the isotherms are generally parallel to the latitude. The deviation from this general trend is more pronounced in January than in July, especially in the Northern Hemisphere. In the Northern Hemisphere the land surface area is much larger than in the Southern Hemisphere. Hence, the effects of land mass and the ocean currents are well pronounced. In January the isotherms deviate to the north over the ocean and to the south over the continent. This can be seen in the North Atlantic Ocean. The presence of warm ocean currents, such as the Gulf Stream and North Atlantic Drift, makes the Northern Atlantic Ocean warmer and the isotherms bend towards the north. Over the land the temperature decreases sharply and the isotherms bend towards south in Europe. Hence, inference 1 is correct while inference 2 is incorrect.